COMPLETE LEARNING SESSION

Growth, Development, HDI, IHDI and MPI - Complete Learning Session

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH

CONCEPT	ARGUMENT	CRITIQUE	APPLICATION
Define precisely	Build the case	Test the limits	Write the answer

Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

Contents

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Growth, Development, HDI, IHDI and MPI - Complete Learning Session

Subject: Economy | **UPSC:** Prelims, GS-II and GS-III | **Current-source cutoff:** 9 September 2026

SOURCE AND DATE CONTROL

Layer	Sources checked	Use in this package
Canonical Core	`upsc-ai-kit\knowledge\Economy\basic\02_Growth-Development-HDI-IHDI-and-MPI.md`	Complete syllabus spine, Indian applications and routed PYQs
Optional Advanced	`upsc-ai-kit\knowledge\Economy\advanced\02_Growth-Development-HDI-IHDI-and-MPI.md`	Separately labelled enrichment only
OCR books	Ramesh Singh, *Indian Economy*, local PDF pp. 96-103; *Economic Survey 2025-26*, especially chapters 11-13	Growth-development distinction, human-capital feedback and dated Indian policy context
UNDP	*Human Development Report 2025 Technical Notes 1-2* and UNDP HDI/IHDI pages, checked 9 September 2026	Exact HDI goalposts, formulae, logarithmic income treatment and IHDI method
OPHI/UNDP	OPHI global MPI methodology page and *Global MPI 2025*, checked 9 September 2026	Ten indicators, weights, dual cutoff, H, A and poverty thresholds
India official	NITI Aayog, *National Multidimensional Poverty Index: A Progress Review 2023*, released 17 July 2023, based on NFHS-4 (2015-16) and NFHS-5 (2019-21)	National MPI architecture, indicator adaptation and state/district use

Vintage rule: A report's publication year, an indicator's reference/data year and a country's rank/value are different facts. This package uses no unsourced current rank or score. Where a dated value appears, its source and reference period are stated together.

LEARNING CONTRACT

Rule	Application
Visual first	Every Core session begins with a topic-specific diagram or comparison.
Core independence	Basic teaching is answer-complete before Optional Advanced depth.
Answer method	Every session uses Claim -> named evidence -> analysis -> qualification.
Data discipline	Every current value states publisher, release/report year and reference period.
Measure discipline	HDI, IHDI, global MPI, national MPI and monetary poverty are not conflated.

BASIC LEARNING SESSION

SESSION 1 - Growth and development: quantity, quality and capabilities

VISUAL FIRST

```
ECONOMIC GROWTH                         ECONOMIC DEVELOPMENT
real output / real income rises           growth plus transformation
              |                                     |
              v                                     v
resource envelope expands                 health + education + agency
                                                 + distribution
                                                 + sustainability

Growth is a means -----------------> Development is the wider end
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

Economic growth is a sustained quantitative increase in real output or real income. Economic development is the broader process through which growth is accompanied by structural change, wider capabilities, fairer distribution, resilience and environmental sustainability.

ANSWER-GRABBING LINE

> Growth enlarges the resource envelope; development judges whether that envelope is converted into productive, equitable and sustainable freedoms.

MUST-WRITE KEYWORDS

- **real GDP**
- **real per-capita income**
- **structural change**
- **capabilities**
- **distribution**
- **sustainability**

CORE EXPLANATION

Nominal output can rise merely because prices rise, so growth comparisons require real measures. Aggregate real GDP can also increase while real GDP per person stagnates if population grows equally fast. Even real per-capita growth remains an average: it says nothing by itself about jobs, access to public services, inequality, unpaid work, security or ecological damage.

Development therefore adds qualitative and distributive questions. Has labour moved from low-productivity to higher-productivity activity? Have nutrition, health, learning and agency improved? Are gains shared across gender, caste, tribe, region and disability? Can the production path continue without eroding the natural and social foundations of future welfare?

Amartya Sen's capability approach supplies the conceptual bridge. Income and commodities are means; the relevant end is the substantive freedom to be healthy, educated, mobile, secure and able to participate. People with identical incomes may convert resources into capabilities differently because public services, disability, discrimination, location and household norms differ.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** Growth and development are related but not interchangeable.
- **Named evidence:** Ramesh Singh's local chapter distinguishes quantitative growth from quantitative-plus-qualitative development; UNDP's human-development framework evaluates health, knowledge and living standards rather than output alone.
- **Analysis:** Real growth can finance private consumption and public revenue, but conversion institutions determine whether resources become capabilities.
- **Qualification:** Growth is normally necessary for durable fiscal and employment capacity, yet neither a high rate nor a high per-capita average proves inclusion or sustainability.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	Ramesh Singh's local chapter distinguishes quantitative growth from quantitative-plus-qualitative development; UNDP's human-development framework evaluates health, knowledge and living standards rather than output alone.
Prelims trap	Do not treat nominal GDP, total GDP or per-capita income as a complete development verdict.
Mains use	Open a growth-versus-development answer with the means-end distinction, then test structure, distribution, capabilities and sustainability.

MINI RECAP

- Growth is quantitative.
- Development is quantitative, structural, distributive and capability-centred.
- Per-capita averages can hide unequal conversion.

SESSION 2 - Drivers of growth and the quality of structural transformation

VISUAL FIRST

```
CAPITAL + LABOUR + HUMAN CAPITAL + TECHNOLOGY + INSTITUTIONS
                              |
                              v
                    PRODUCTIVITY GROWTH
                              |
                  +-----------+-----------+
                  |                       |
                  v                       v
        HIGHER REAL OUTPUT         STRUCTURAL CHANGE
                                   agriculture -> industry/services
                  |                       |
                  +-----------+-----------+
                              v
               JOBS + WAGES + TAX BASE + CAPABILITIES
                              |
                  quality depends on inclusion and ecology
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

Growth accounting links output expansion to labour, physical capital, human capital and productivity; structural transformation reallocates workers and resources toward more productive activities.

ANSWER-GRABBING LINE

> The development content of growth lies not only in its speed but in its productivity source, employment intensity, sectoral composition and ecological cost.

MUST-WRITE KEYWORDS

- **capital accumulation**
- **labour productivity**
- **total factor productivity**
- **human capital**
- **structural transformation**
- **jobless growth**

CORE EXPLANATION

Physical investment expands productive capacity, while education, health and skills improve the quality of labour. Technology, infrastructure, competition and capable institutions raise the efficiency with which factors are combined. Sustained per-capita growth ultimately depends heavily on productivity rather than indefinite factor accumulation.

Structural transformation is development-enhancing when workers leave low-productivity, vulnerable activity for higher-productivity work with better wages, security and learning. A shift in output shares without adequate movement into decent jobs can produce job-poor or jobless growth. Capital-intensive enclaves may lift GDP while weakly spreading incomes.

Inclusive growth combines participation in production with access to opportunities and a fair distribution of gains. Sustainable growth adds resource efficiency, pollution control, climate resilience and protection of natural capital. A short-run output gain created by depletion or unpriced environmental damage can reduce future capability.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** A growth rate becomes developmentally meaningful only after its source and employment transmission are examined.
- **Named evidence:** *Economic Survey 2025-26*, chapters 11-12, connects education, health and skills with productivity, employability, social mobility and structural transformation.
- **Analysis:** Human capital raises productivity and earnings; productive jobs then finance household investment and widen the tax base, creating a feedback loop.
- **Qualification:** Employment quantity is insufficient if work is informal, unsafe, low-paid or environmentally destructive.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	*Economic Survey 2025-26*, chapters 11-12, connects education, health and skills with productivity, employability, social mobility and structural transformation.
Prelims trap	Structural transformation means a productivity-enhancing reallocation, not merely a falling agricultural share in GDP.
Mains use	Use a four-link chain: productivity source -> sector/jobs -> distribution/public revenue -> capability and environmental outcome.

MINI RECAP

- Capital deepening and productivity drive output.
- Jobs connect growth to household welfare.
- Sustainability protects future development.

SESSION 3 - HDI purpose, timeline and exact architecture

VISUAL FIRST

```
1990: FIRST UNDP HUMAN DEVELOPMENT REPORT
                   |
                   v
            HUMAN DEVELOPMENT INDEX
       +-----------+-----------+-----------+
       |                       |           |
       v                       v           v
 LONG AND HEALTHY LIFE      KNOWLEDGE    DECENT LIVING STANDARD
 life expectancy at birth   MYS + EYS    GNI per capita, PPP
       |                       |           |
       +----------- normalized indices ---+
                           |
                           v
            HDI = geometric mean of 3 dimensions
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

The Human Development Index is the United Nations Development Programme (UNDP) summary measure of average achievement in three dimensions: a long and healthy life, access to knowledge and a decent standard of living.

ANSWER-GRABBING LINE

> HDI shifts comparison from the size of an economy to the average capabilities that income, institutions and public action help create.

MUST-WRITE KEYWORDS

- **UNDP**
- **1990 HDR**
- **three dimensions**
- **four indicators**
- **normalisation**
- **geometric mean**

CORE EXPLANATION

The health dimension uses life expectancy at birth. Knowledge uses two indicators: expected years of schooling for a child of school-entering age and mean years of schooling for adults aged 25 and above. The standard-of-living dimension uses gross national income per capita expressed in purchasing-power-parity terms. Thus HDI has three dimensions but four component indicators.

Each indicator is converted into a dimension index between zero and one. The two schooling indices are first calculated separately and then averaged arithmetically to form the education index. The health, education and income dimension indices are then aggregated by geometric mean.

HDI is designed to provoke policy comparison. Countries with similar GNI per capita can have different HDI outcomes because public health, education, distribution and institutions affect conversion. It is not a measure of total welfare, happiness or political freedom.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** HDI is an average capability-achievement index, not an income ranking with social variables attached.
- **Named evidence:** UNDP's 2025 Technical Note 1 defines the three dimensions, four indicators and geometric-mean architecture.
- **Analysis:** The index makes health and education co-equal with income and limits the ability of one very high dimension to compensate for another very low one.
- **Qualification:** Its narrow dimensions and national averages still omit inequality, poverty, security, empowerment and ecological pressures.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	UNDP's 2025 Technical Note 1 defines the three dimensions, four indicators and geometric-mean architecture.
Prelims trap	HDI uses GNI per capita at PPP, not GDP alone, and expected plus mean years of schooling, not literacy alone.
Mains use	State purpose first, decode all four indicators, explain geometric aggregation and end with distributional limits.

MINI RECAP

- Three dimensions, four indicators.
- Education averages MYS and EYS indices.
- UNDP aggregates dimensions geometrically.

SESSION 4 - HDI calculation: goalposts, logarithmic income and a worked example

VISUAL FIRST

```
NORMALISE EACH INDICATOR
index = (actual - minimum) / (maximum - minimum)

HEALTH: LE 20 to 85
EDUCATION: EYS 0 to 18; MYS 0 to 15; average the two indices
INCOME: GNI pc 100 to 75,000 in constant 2021 PPP dollars
        use natural logarithms

HDI = (Health index x Education index x Income index)^(1/3)
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

Goalposts are the minimum and maximum values used by the applicable UNDP technical note to normalise indicators expressed in different units.

ANSWER-GRABBING LINE

> HDI first makes unlike indicators comparable, then applies diminishing returns to income and geometric aggregation across dimensions.

MUST-WRITE KEYWORDS

- **goalposts**
- **life expectancy index**
- **education index**
- **income index**
- **natural logarithm**
- **2021 PPP dollars**

CORE EXPLANATION

The applicable *Human Development Report 2025 Technical Note 1* sets these goalposts: life expectancy 20-85 years; expected years of schooling 0-18; mean years of schooling 0-15; and GNI per capita 100-75,000 in constant 2021 PPP dollars. Goalposts belong to a methodology vintage and should not be copied into another report year without checking.

For health and each schooling indicator, use `(actual - minimum)/(maximum - minimum)`. Education is `(EYS index + MYS index)/2`. Income is `[ln(actual GNIpc) - ln(100)]/[ln(75,000) - ln(100)]`; logarithms reflect diminishing capability gains from additional income.

Illustrative calculation, not a country estimate: life expectancy 72, EYS 14, MYS 10 and GNI per capita 20,000 produce health `52/65 = 0.800`; EYS `14/18 = 0.778`; MYS `10/15 = 0.667`; education `0.722`; income approximately `0.800`. HDI is approximately `(0.800 x 0.722 x 0.800)^(1/3) = 0.773`.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** The logarithmic income index prevents equal dollar increments from being treated as equal capability gains at every income level.
- **Named evidence:** UNDP 2025 Technical Note 1 explicitly uses natural logarithms and constant 2021 PPP GNI per capita with a 100-75,000 goalpost range.
- **Analysis:** An additional unit of income is assumed to matter more near subsistence than at high income, while PPP improves cross-country purchasing-power comparability.
- **Qualification:** PPP estimates, input revisions and goalpost choices affect measured values; the worked calculation is pedagogical, not an official country value.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	UNDP 2025 Technical Note 1 explicitly uses natural logarithms and constant 2021 PPP GNI per capita with a 100-75,000 goalpost range.
Prelims trap	Do not use a simple arithmetic mean of the three final dimensions or a linear income index.
Mains use	Write the formula and one line on why logarithms and the geometric mean are used; calculations can secure Prelims elimination.

MINI RECAP

- Check the technical-note vintage.
- Income is log-transformed.
- HDI is the cube root of the product.

SESSION 5 - Reading HDI correctly: value, rank, data year and limitations

VISUAL FIRST

```
REPORT YEAR != DATA YEAR != RANK != VALUE

HDI value: achievement on 0-1 scale under stated method
Rank: ordinal position among countries included in that release
Data year: reference year of component observations
Report year: publication label/date

Small value revision + changed country coverage -> rank may move without a comparable welfare jump
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

An HDI value is the composite score; an HDI rank is the country's relative order in a stated release and comparison set.

ANSWER-GRABBING LINE

> A rank is a relative position, not a welfare unit: serious analysis compares values, components, vintages and uncertainty before celebrating movement.

MUST-WRITE KEYWORDS

- **value**
- **rank**
- **data year**

- **report year**
- **revision**
- **comparability**

CORE EXPLANATION

A report published in one year commonly reports indicators referring mainly to an earlier year. Component indicators may also come from different statistical sources and revision cycles. Therefore write, for example, 'the 2025 report's value for reference year 2023' rather than calling it the '2025 level' without explanation.

Rank depends on other countries, data availability and revisions. A country can improve its value yet lose rank if peers improve faster; it can gain rank with little domestic change. Compare a consistent time series supplied by UNDP rather than subtracting ranks from differently constituted releases.

HDI is intentionally parsimonious. It omits within-country distribution, multidimensional poverty, political voice, safety, unpaid care, service quality and environmental pressure. The Gender Development Index (GDI) compares female and male HDI achievements; the Gender Inequality Index (GII) combines reproductive health, empowerment and labour-market dimensions. They are bounded complements, not fourth HDI dimensions.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** Composite indices are navigation aids, not complete social accounts.
- **Named evidence:** UNDP's HDI page states that HDI captures only part of human development and does not itself reflect inequalities, poverty, human security or empowerment.
- **Analysis:** A dashboard of component levels and disaggregated outcomes preserves diagnosis that a single score or rank suppresses.
- **Qualification:** Comparability is valuable, but simplification inevitably trades detail for communicability.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	UNDP's HDI page states that HDI captures only part of human development and does not itself reflect inequalities, poverty, human security or empowerment.
Prelims trap	Never mix report year, underlying data year and rank; never call GDI or GII an HDI dimension.
Mains use	After explaining HDI, add a two-column value-versus-rank distinction and recommend a dashboard.

MINI RECAP

- Value is cardinal-like; rank is ordinal.
- Data vintage must be named.
- GDI/GII complement rather than enlarge HDI.

SESSION 6 - IHDI: inequality discount, dimension losses and equality case

VISUAL FIRST

```
HDI DIMENSION INDEX Ix
        |
        +-- estimate inequality Ax in that dimension
        |
        v
ADJUSTED INDEX I-adjusted = (1 - Ax) x I

IHDI = (Health-adjusted x Education-adjusted x Income-adjusted)^(1/3)
Loss = 1 - (IHDI / HDI)

perfect equality -> every Ax = 0 -> IHDI = HDI
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

The Inequality-adjusted Human Development Index discounts each HDI dimension according to inequality in its distribution and then geometrically aggregates the adjusted indices.

ANSWER-GRABBING LINE

> HDI reports the average frontier; IHDI asks how much of that achievement is lost when unequal distribution is recognised.

MUST-WRITE KEYWORDS

- **Atkinson inequality**
- **dimension-level discount**
- **IHDI <= HDI**
- **loss due to inequality**
- **equality case**
- **distribution-sensitive**

CORE EXPLANATION

UNDP's 2025 Technical Note 2 draws on the Atkinson family with inequality-aversion parameter epsilon equal to one. For a dimension, `A = 1 - geometric mean/arithmetic mean`. The adjusted dimension index is `(1-A) x I`. The IHDI is the geometric mean of the three adjusted indices.

The proportional loss due to inequality can be written `1 - IHDI/HDI`. If health, education and income are equally distributed, every inequality discount is zero and IHDI equals HDI. As inequality rises, IHDI falls below HDI. IHDI is therefore not a fourth dimension and the HDI-IHDI gap is not a poverty rate.

The distributions use different units and data sources: life expectancy across a hypothetical cohort, schooling across individuals and income/consumption across individuals. UNDP notes that IHDI is not association-sensitive because all dimensions are not observed jointly for the same individuals in a single survey across countries.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** IHDI is better suited than HDI to an inclusive-growth question because it penalises unequal achievement within the same three dimensions.
- **Named evidence:** UNDP 2025 Technical Note 2 defines dimension-specific Atkinson discounts, the geometric aggregation and the equality case.
- **Analysis:** Two countries with the same HDI can have different IHDIs when one distributes longevity, schooling or income more unequally.
- **Qualification:** IHDI still does not reveal which people suffer simultaneous disadvantages across dimensions, and data coverage is narrower than HDI.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	UNDP 2025 Technical Note 2 defines dimension-specific Atkinson discounts, the geometric aggregation and the equality case.
Prelims trap	IHDI does not add inequality as a fourth dimension; it adjusts each existing dimension.
Mains use	For HDI versus IHDI, compare purpose, formula, distribution sensitivity, equality case and residual limits.

MINI RECAP

- Adjust all three dimensions.
- IHDI equals HDI only at equality.
- Loss is proportional, not a poverty headcount.

SESSION 7 - Global MPI: dimensions, ten indicators, weights and dual cutoff

VISUAL FIRST

```
GLOBAL MPI (OPHI + UNDP)
HEALTH 1/3             EDUCATION 1/3          LIVING STANDARDS 1/3
nutrition 1/6          years schooling 1/6    fuel, sanitation, water,
child mortality 1/6    attendance 1/6         electricity, housing, assets
                                               each 1/18
                 household deprivation score c
                             |
                c >= 1/3 -> MPI poor
             1/5 <= c < 1/3 -> vulnerable
                c >= 1/2 -> severe poverty
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

The global MPI, developed by the Oxford Poverty and Human Development Initiative (OPHI) with the United Nations Development Programme (UNDP), identifies acute multidimensional poverty through simultaneous weighted deprivations in health, education and living standards using the Alkire-Foster method.

ANSWER-GRABBING LINE

> The MPI preserves both identification and aggregation: it first identifies who crosses a deprivation cutoff, then records how many are poor and how intensely they are deprived.

MUST-WRITE KEYWORDS

- **OPHI**
- **UNDP**
- **Alkire-Foster**
- **ten indicators**
- **deprivation cutoff**
- **poverty cutoff**

CORE EXPLANATION

The global MPI has ten indicators. Health: nutrition and child mortality, each weight 1/6. Education: years of schooling and school attendance, each 1/6. Living standards: cooking fuel, sanitation, drinking water, electricity, housing and assets, each 1/18. Each dimension therefore contributes one-third.

The first cutoff is indicator-specific: it decides whether the household is deprived on each indicator. Weighted deprivations are then summed into score `c`. The second, poverty cutoff identifies a person as MPI poor when the household score is at least one-third.

Under the methodology checked on 9 September 2026, a score from one-fifth up to but below one-third is classified as vulnerable to multidimensional poverty; one-half or more is severe multidimensional poverty. These thresholds belong to the current global method and must not be transferred automatically to a national MPI.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** The global MPI measures overlapping acute deprivation rather than low income alone.
- **Named evidence:** OPHI's current global MPI methodology page specifies the three dimensions, ten indicators, weights and one-third poverty cutoff; the 2025 global report continues this architecture.
- **Analysis:** A household-level score reveals whether deprivations cluster, enabling indicator and subgroup decomposition.
- **Qualification:** Household identification can attribute a household deprivation to every member and thereby miss intra-household differences.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	OPHI's current global MPI methodology page specifies the three dimensions, ten indicators, weights and one-third poverty cutoff; the 2025 global report continues this architecture.
Prelims trap	Equal dimension weights do not mean all ten indicators have equal weights.
Mains use	Draw the 2+2+6 indicator tree and explain the two cutoffs before using H and A.

MINI RECAP

- Three equal dimensions.
- Ten unequally weighted indicators.
- One-third is the global poverty cutoff.

SESSION 8 - MPI aggregation: incidence H, intensity A and worked calculation

VISUAL FIRST

```
FIVE EQUAL-SIZED HOUSEHOLDS: deprivation scores
0.50 | 0.40 | 0.35 | 0.25 | 0.10
 poor   poor   poor   not    not       (cutoff = 1/3)

H = 3/5 = 0.60
A = (0.50 + 0.40 + 0.35) / 3 = 0.4167
MPI = H x A = 0.250

H asks HOW MANY; A asks HOW DEPRIVED among the poor.
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

Incidence `H` is the population share identified as multidimensionally poor; intensity `A` is the average weighted deprivation score among the poor; `MPI = H x A`.

ANSWER-GRABBING LINE

> A poverty reduction claim is incomplete until it states whether fewer people are poor, the remaining poor are less deprived, or both.

MUST-WRITE KEYWORDS

- **censored headcount**
- **incidence H**
- **intensity A**
- **MPI = H x A**

- **decomposition**
- **cutoff sensitivity**

CORE EXPLANATION

After applying the poverty cutoff, deprivations of non-poor households are censored to zero for aggregation. `H = q/n`, where q is the number of poor persons and n the population. `A` is the mean deprivation score among q. Their product is the adjusted headcount ratio or MPI.

In the example, three of five equal-sized households are poor, so H is 0.60. Their average score is 0.4167, producing MPI 0.250. If the 0.35 household moves just below one-third, H falls sharply even if the two poorest households do not improve. If the poorest households lose some deprivations but remain poor, A falls while H is unchanged.

MPI can be decomposed by subgroup and indicator, helping identify whether a district's burden arises mainly from nutrition, schooling or amenities. But a single cross-section does not prove that the same household escaped poverty; repeated cross-sections show aggregate change, not individual trajectories.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** H and A prevent a headcount-only policy from ignoring the depth and composition of multidimensional poverty.
- **Named evidence:** The Alkire-Foster aggregation used by OPHI/UNDP explicitly reports incidence and intensity and multiplies them.
- **Analysis:** Because MPI changes when either the number poor or their deprivation share changes, it rewards improvements below the poverty cutoff as well as exits.
- **Qualification:** Results remain sensitive to indicators, deprivation cutoffs, weights, poverty cutoff and survey error.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	The Alkire-Foster aggregation used by OPHI/UNDP explicitly reports incidence and intensity and multiplies them.
Prelims trap	Intensity is calculated among the multidimensionally poor, not across the whole population.
Mains use	Use one numerical example, then explain how H and A can move differently.

MINI RECAP

- H is a proportion of people.
- A is the poor's average deprivation share.
- MPI responds to both incidence and intensity.

SESSION 9 - India's National MPI: adapted architecture and official evidence

VISUAL FIRST

```
NITI AAYOG NATIONAL MPI
NFHS household microdata -> indicator deprivation -> weighted score

HEALTH (1/3)         EDUCATION (1/3)      STANDARD OF LIVING (1/3)
nutrition 1/6        years schooling 1/6  fuel, sanitation, water,
child/adolescent     attendance 1/6       electricity, housing, assets,
mortality 1/12                             bank account - each 1/21
maternal health 1/12

national -> State/UT -> district diagnosis
NFHS-4 (2015-16) compared with NFHS-5 (2019-21) in 2023 report
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

India's National MPI is the National Institution for Transforming India (NITI Aayog) nationally adapted Alkire-Foster measure based principally on National Family Health Survey (NFHS) household microdata.

ANSWER-GRABBING LINE

> India's national MPI retains a comparable multidimensional logic but adapts indicators and weights to domestic policy needs; it is not the global MPI with an India label.

MUST-WRITE KEYWORDS

- **NITI Aayog**
- **NFHS-4**
- **NFHS-5**
- **12 indicators**
- **maternal health**
- **bank account**
- **district estimates**

CORE EXPLANATION

The 2023 Progress Review uses 12 indicators across the same three one-third dimensions. Health weights are nutrition 1/6, child and adolescent mortality 1/12 and maternal health 1/12. Education has years of schooling and school attendance, each 1/6. Standard of living has seven indicators - cooking fuel, sanitation, drinking water, electricity, housing, assets and bank account - each 1/21.

NITI Aayog is the nodal institution for the national MPI exercise; NFHS is conducted by the International Institute for Population Sciences under the Ministry of Health and Family Welfare framework. The index supports national, State/UT and district diagnosis, indicator decomposition and targeted action.

The report released 17 July 2023 compares NFHS-4 (2015-16) with NFHS-5 (2019-21). It reports India's multidimensional-poverty headcount ratio falling from 24.85 percent to 14.96 percent and estimates 13.5 crore people moved out of multidimensional poverty over that comparison. These are report facts tied to survey periods, not 2023 observations.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** National adaptation increases policy relevance while reducing direct interchangeability with the global MPI.
- **Named evidence:** NITI Aayog's 2023 report specifies the 12-indicator national architecture and uses NFHS-4 and NFHS-5 for state and district comparison.
- **Analysis:** Maternal health and bank-account access add Indian policy salience, while disaggregation helps locate the dimensions and places requiring action.
- **Qualification:** NFHS periodicity creates data lag; state/district averages and household identification do not expose every intra-household deprivation.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	NITI Aayog's 2023 report specifies the 12-indicator national architecture and uses NFHS-4 and NFHS-5 for state and district comparison.
Prelims trap	Do not quote the national MPI's 12 indicators or weights as the global MPI's ten-indicator method.
Mains use	Name NITI Aayog, NFHS vintages, 12-indicator adaptation and district use, then add comparability and lag cautions.

MINI RECAP

- National MPI is adapted, not identical.
- NFHS supplies household data.
- The 2023 report compares 2015-16 with 2019-21.

SESSION 10 - Income poverty, multidimensional deprivation and measurement limits

VISUAL FIRST

```
MONETARY POVERTY                        MULTIDIMENSIONAL POVERTY
income/consumption < line                weighted deprivations >= cutoff
command over market goods               direct capability failures
        |                                        |
        +--------------- COMPLEMENTS -------------+
                         |
 public services, prices, household needs and location can make them diverge
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

A monetary poverty measure identifies insufficient income or consumption relative to a line; MPI identifies simultaneous direct deprivations relative to weighted indicator and poverty cutoffs.

ANSWER-GRABBING LINE

> Income poverty asks whether resources cross a monetary line; MPI asks whether functionings fail together, so neither measure can safely replace the other.

MUST-WRITE KEYWORDS

- **poverty line**
- **headcount ratio**
- **direct deprivation**
- **aggregation**
- **cutoff sensitivity**
- **intra-household blindness**

CORE EXPLANATION

Income or consumption measures are indispensable for assessing purchasing power, transfer adequacy and macroeconomic shocks. Yet the same income can produce different outcomes when public health, schooling, water, prices, disability or household composition differ. MPI observes selected outcomes directly and reveals overlap.

Both approaches make normative and statistical choices. A poverty line fixes a monetary threshold and price adjustment. MPI fixes dimensions, indicators, deprivation cutoffs, weights and an overall poverty cutoff. Small movement around either threshold can change classification, although continuous measures and sensitivity checks can reduce over-interpretation.

The household is commonly the global and national MPI identification unit. If one child is out of school or a member is undernourished, all household members may inherit that indicator deprivation. This supports household targeting but can conceal unequal allocation by gender, age or disability. Survey periodicity, recall, missing data and data lag further constrain current claims.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** Poverty measurement should triangulate monetary command, direct deprivations and subgroup evidence.
- **Named evidence:** The 2020 GS-II routed demand explicitly contrasts incidence and intensity with income-based measurement; NITI's national MPI is designed as a complementary diagnostic.
- **Analysis:** A combined dashboard distinguishes shortage of money from failures of service access, quality and household conversion.
- **Qualification:** More dimensions do not automatically mean a better index; indicator validity, transparency and robustness matter.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	The 2020 GS-II routed demand explicitly contrasts incidence and intensity with income-based measurement; NITI's national MPI is designed as a complementary diagnostic.
Prelims trap	MPI is not a monetary poverty line, and monetary headcount is not MPI incidence.
Mains use	Compare unit, threshold, information revealed, policy use and blind spots in a compact matrix.

MINI RECAP

- Monetary and multidimensional poverty differ.
- Threshold choices matter in both.
- Household measures can hide individual inequality.

SESSION 11 - Growth-development feedback, jobless growth and policy design

VISUAL FIRST

```
PRODUCTIVITY / REAL GROWTH
        |
        +--> jobs and wages ------> nutrition, schooling, security
        |
        +--> profits/investment --> technology and capacity
        |
        '--> tax base -----------> public health, education, infrastructure
                                      |
                                      v
                              HUMAN CAPITAL / AGENCY
                                      |
                                      +----> productivity feedback

FAILURE MODES: jobless growth | inequality | weak services | pollution | climate risk
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

The growth-development feedback loop links productivity and output to jobs, incomes and fiscal capacity, then links health, education and security back to labour productivity and innovation.

ANSWER-GRABBING LINE

> Human development is not merely a distributional use of growth; it is productive infrastructure that can strengthen the quality and durability of future growth.

MUST-WRITE KEYWORDS

- **virtuous cycle**
- **employment elasticity**
- **human-capital feedback**
- **social protection**
- **productive inclusion**
- **just transition**

CORE EXPLANATION

Growth can reduce deprivation through jobs, wages, cheaper goods and public revenue. The channel weakens when output is concentrated in capital-intensive sectors, labour participation is constrained, wages lag productivity or public services are inaccessible. This is the jobless-growth problem: output rises without commensurate decent employment.

Inequality can obstruct capability formation by limiting nutrition, schooling, credit and voice; poor human capital then depresses productivity and mobility. Social protection cushions shocks and prevents distress sales or school withdrawal, while quality health, education, skills and infrastructure build productive capability. Protection and transformation are complements.

Environmental trade-offs require more than attaching a green adjective. Pollution harms health and labour productivity; resource depletion shifts costs to future generations; climate shocks can reverse poverty exits. Policy should support employment-intensive productivity, universal quality basic services, targeted support for residual deprivation, regional investment and a just transition.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** Inclusive development requires both productive participation and protection against capability-destroying shocks.
- **Named evidence:** *Economic Survey 2025-26* links education, health, skilling and employment to productivity and social mobility; NITI MPI enables deprivation-specific territorial targeting.
- **Analysis:** Public action can convert fiscal resources into human capital, which expands both welfare and future productive capacity.
- **Qualification:** Higher expenditure is an input; access, quality, utilisation and outcomes must be separately demonstrated.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	*Economic Survey 2025-26* links education, health, skilling and employment to productivity and social mobility; NITI MPI enables deprivation-specific territorial targeting.
Prelims trap	Inclusive growth is not synonymous with redistribution after growth or with welfare spending alone.
Mains use	Organise policy under productive jobs, universal capabilities, targeted protection, territorial convergence and environmental resilience.

MINI RECAP

- Jobs transmit growth.
- Capabilities feed productivity.
- Expenditure must be tested against outcomes.

SESSION 12 - Comparison dashboard: HDI, IHDI, global MPI and national MPI

VISUAL FIRST

```
QUESTION                  BEST STARTING MEASURE
Average achievement       HDI
Achievement after         IHDI
inequality
Acute overlapping         Global MPI
deprivation, comparable
Domestic State/district   India's National MPI
policy diagnosis

No single index answers all four questions.
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

A measurement dashboard assigns each index to the question it was designed to answer and prevents cross-index substitution.

ANSWER-GRABBING LINE

> Choose the metric after defining the policy question: averages, distribution, acute overlap and domestic targeting are distinct analytical tasks.

MUST-WRITE KEYWORDS

- **average achievement**
- **distribution adjustment**
- **acute deprivation**
- **national adaptation**
- **dashboard**
- **decomposition**

CORE EXPLANATION

HDI is country-level average achievement in three dimensions. IHDI retains those dimensions but discounts them for inequality. Global MPI identifies acute household deprivations using ten internationally comparable indicators. India's national MPI uses 12 nationally adapted indicators and NFHS data for domestic diagnosis.

HDI and IHDI use achievement indices; MPI uses deprivation indicators and a dual-cutoff method. IHDI's inequality adjustment is not the same as MPI's identification of overlapping deprivation. Global and national MPI values are not interchangeable because indicator definitions and weights differ.

For Prelims, decode publisher, unit, dimensions, indicators, formula and threshold. For Mains, use a dashboard: real per-capita growth, employment and productivity; HDI components; IHDI loss; monetary poverty; MPI H, A and contributions; gender and regional disaggregation; environmental indicators.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** Measurement pluralism is disciplined when each index has a defined purpose rather than being treated as competing truth claims.
- **Named evidence:** UNDP, OPHI and NITI Aayog publish distinct architectures for average achievement, inequality adjustment and multidimensional deprivation.
- **Analysis:** Using complementary measures reduces the risk that progress in one aggregate masks failure in another dimension or group.
- **Qualification:** A larger dashboard increases information but also demands consistent vintages, transparent metadata and careful communication.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	UNDP, OPHI and NITI Aayog publish distinct architectures for average achievement, inequality adjustment and multidimensional deprivation.
Prelims trap	HDI, IHDI and MPI cannot be ranked as broader and narrower versions of one formula.
Mains use	Use the measure-purpose matrix as the analytical core and finish with disaggregated dashboard governance.

MINI RECAP

- Match measure to question.
- Do not mix architectures.
- Metadata is part of the answer.

SESSION 13 - Exam synthesis: calculation, interpretation and qualified policy verdict

VISUAL FIRST

```
DEFINE THE QUESTION
      |
      v
SELECT METRIC -> decode institution + unit + formula + threshold
      |
      v
READ RESULT -> value + component + distribution + data year
      |
      v
EXPLAIN MECHANISM -> growth/jobs/services/institutions/environment
      |
      v
QUALIFY -> averages + cutoffs + household unit + lag + causation
      |
      v
POLICY -> productive inclusion + capabilities + resilience
```

Visual purpose: establish the decision path before introducing prose.

DEFINITION

Exam synthesis connects a correctly decoded metric to an economic mechanism, named evidence, measurement qualification and feasible policy response.

ANSWER-GRABBING LINE

> The examiner rewards not the largest data dump but the cleanest chain from concept to measurement, mechanism, evidence, limitation and policy.

MUST-WRITE KEYWORDS

- **decode**
- **calculate**
- **interpret**
- **disaggregate**
- **qualify**
- **policy feedback**

CORE EXPLANATION

For a calculation question, state the unit and apply the exact formula. For HDI, normalise indicators, average the two schooling indices, log-transform income and use the geometric mean. For MPI, identify indicator deprivations, apply weights and the poverty cutoff, censor non-poor deprivations, calculate H and A, then multiply.

For an interpretive question, never infer a welfare story from rank alone. State publisher, release/report year, data/reference year and whether the value is a survey observation, estimate or projection. Compare like with like.

For a policy question, distinguish growth creation from conversion. Productivity, investment and structural transformation supply resources; jobs, health, education, institutions and environmental resilience determine distribution and durability. Close by recommending a transparent, disaggregated dashboard rather than replacing GDP with one composite index.

CLAIM -> NAMED EVIDENCE -> ANALYSIS -> QUALIFICATION

- **Claim:** A strong development answer treats measurement as part of causal analysis, not as a decorative statistic.
- **Named evidence:** The 2025 GS-III question directly requires HDI-IHDI distinction for inclusive growth, while the 2020 GS-II demand requires H-A decomposition against income poverty.
- **Analysis:** Formula literacy prevents conceptual errors; mechanism and qualification convert description into evaluation.
- **Qualification:** No index can establish causation without policy, institutional and counterfactual evidence.

EVIDENCE, PRELIMS TRAP AND MAINS USE

Exam tool	Topic-specific use
Evidence	The 2025 GS-III question directly requires HDI-IHDI distinction for inclusive growth, while the 2020 GS-II demand requires H-A decomposition against income poverty.
Prelims trap	Do not quote an undated rank, threshold or percentage; do not treat a publication year as the observation year.
Mains use	Use the answer spine: define -> formula -> evidence -> mechanism -> limitation -> targeted and sustainable policy.

MINI RECAP

- Decode before comparing.
- Attach dates to statistics.
- Conclude with capabilities, inclusion and sustainability.

Practice contract: Exactly 32 original MCQs appear before PYQs. Correct answers follow `ABCD` repeated eight times. Every option receives a substantive question-specific explanation and every question ends with a unique examiner trap.

BASIC MCQS / REMEDIATION

MCQ 1

Which observation is sufficient by itself to establish economic growth in the strict sense?

- A. Nominal GDP rose while the price level also rose
- B. Real GDP increased over the stated period
- C. The HDI rank improved
- D. Public social expenditure increased

Answer: A.

Option-specific explanations:

- **A - Correct:** Nominal expansion can be entirely price-driven, so it does not establish higher real output.
- **B - Incorrect:** A rise in real GDP directly records a quantitative expansion of inflation-adjusted output.
- **C - Incorrect:** HDI rank is a relative human-development position, not a growth measure.

- **D - Incorrect:** Expenditure is an input and may rise without measured output growth.

Examiner trap 1: Growth is established by a real quantity measure, not by a welfare or spending proxy.

MCQ 2

Which statement best distinguishes development from growth?

- A. Development excludes changes in output
- B. Development adds structural, distributive, capability and sustainability dimensions to quantitative expansion
- C. Development is measured only by per-capita GNI
- D. Development begins only after growth has ended

Answer: B.

Option-specific explanations:

- **A - Incorrect:** Development normally includes the resource gains produced by growth rather than excluding output.
- **B - Correct:** Development asks how growth changes structure, capabilities, distribution and sustainability.
- **C - Incorrect:** Per-capita GNI is one means and one average, not the sole definition.
- **D - Incorrect:** Growth and development interact; they are not sequential non-overlapping eras.

Examiner trap 2: Do not define development as growth plus any one social indicator.

MCQ 3

A country records rapid real GDP growth but stagnant employment and widening regional deprivation. The most accurate description is

- A. development without growth
- B. negative growth with inclusion
- C. job-poor growth with weak conversion into broad development
- D. proof that productivity has fallen

Answer: C.

Option-specific explanations:

- **A - Incorrect:** Real GDP growth rules out the description of no growth.
- **B - Incorrect:** Stagnant employment and widening deprivation do not establish inclusion.
- **C - Correct:** Output can expand with weak labour absorption and uneven capability conversion.
- **D - Incorrect:** Productivity can rise in capital-intensive sectors even when employment stagnates.

Examiner trap 3: Jobless growth concerns employment transmission, not necessarily absence of productivity.

MCQ 4

Which policy package most directly supports sustainable inclusive growth?

- A. Only raising aggregate investment
- B. Only increasing cash transfers
- C. Only improving the HDI rank
- D. Employment-intensive productivity, quality basic services, protection against shocks and ecological resilience

Answer: D.

Option-specific explanations:

- **A - Incorrect:** Investment matters but its sectoral, labour and environmental effects must be tested.
- **B - Incorrect:** Transfers protect consumption but do not alone create capabilities or productive jobs.
- **C - Incorrect:** A rank target can encourage indicator gaming and ignores mechanisms.
- **D - Correct:** The package combines production, participation, capability, protection and sustainability.

Examiner trap 4: Inclusive growth is both a process of participation and an outcome of shared capability.

MCQ 5

How many formal dimensions and component indicators does the HDI have under UNDP's 2025 method?

- A. Three dimensions and four indicators
- B. Four dimensions and four indicators
- C. Three dimensions and three indicators
- D. Four dimensions and five indicators

Answer: A.

Option-specific explanations:

- **A - Correct:** Health, education and living standard are three dimensions; education has two indicators, making four indicators overall.
- **B - Incorrect:** Inequality is not a fourth HDI dimension.
- **C - Incorrect:** Education uses both mean and expected years, so there are more than three indicators.
- **D - Incorrect:** Political freedom is not an HDI component indicator.

Examiner trap 5: Do not count two education indicators as two dimensions.

MCQ 6

Which is the HDI standard-of-living indicator?

- A. Real GDP per capita at market exchange rates
- B. GNI per capita in constant 2021 PPP dollars under the 2025 technical note
- C. Household consumption below a poverty line
- D. Average household wealth

Answer: B.

Option-specific explanations:

- **A - Incorrect:** GDP and market exchange rates do not match the UNDP definition.
- **B - Correct:** UNDP uses GNI per capita in PPP terms and identifies the PPP/base vintage.
- **C - Incorrect:** A poverty-line headcount belongs to monetary poverty measurement.
- **D - Incorrect:** Wealth is not the formal HDI living-standard indicator.

Examiner trap 6: HDI uses GNI, PPP and a dated methodology, not GDP alone.

MCQ 7

Under the 2025 UNDP technical note, the education index is formed by

- A. multiplying literacy and enrolment rates
- B. using mean years of schooling alone
- C. averaging the normalised expected-years and mean-years schooling indices
- D. taking the geometric mean of school attendance and attainment

Answer: C.

Option-specific explanations:

- **A - Incorrect:** Literacy and enrolment are not the current component pair.
- **B - Incorrect:** Mean years captures adult attainment but not expected schooling for entrants.
- **C - Correct:** Each schooling indicator is normalised, then their indices are averaged arithmetically.
- **D - Incorrect:** The named indicators and aggregation are incorrect.

Examiner trap 7: The final HDI is geometric, but the two education subindices are arithmetically averaged.

MCQ 8

Why is GNI per capita logarithmically transformed in HDI?

- A. To remove all income inequality
- B. To convert nominal income into real income
- C. To make income equal in weight to population
- D. To reflect diminishing capability gains from additional income

Answer: D.

Option-specific explanations:

- **A - Incorrect:** Inequality is addressed by IHDI, not removed by a log.
- **B - Incorrect:** PPP and constant-price procedures address purchasing power and price comparability.
- **C - Incorrect:** Weighting and transformation are different operations.
- **D - Correct:** UNDP treats additional income as contributing less to capabilities at higher levels.

Examiner trap 8: The logarithm is about diminishing returns, not inequality adjustment.

MCQ 9

Which set gives the 2025 HDI goalposts correctly?

- A. Life expectancy 20-85; EYS 0-18; MYS 0-15; GNIpc 100-75,000 in 2021 PPP dollars
- B. Life expectancy 0-100; literacy 0-100; GDPpc 0-100,000
- C. Life expectancy 25-80; EYS 0-15; MYS 0-12; GNIpc 500-50,000
- D. The goalposts equal the lowest and highest country values each year

Answer: A.

Option-specific explanations:

- **A - Correct:** These are the explicit minimum and maximum values in UNDP Technical Note 1 for HDR 2025.
- **B - Incorrect:** The indicators and bounds do not match the current method.
- **C - Incorrect:** These values are not the applicable technical-note goalposts.
- **D - Incorrect:** The method uses fixed stated goalposts rather than automatically adopting each year's extremes.

Examiner trap 9: Goalposts are methodology-specific; always state the report/technical-note vintage.

MCQ 10

What does the geometric mean in HDI primarily do?

- A. Makes every dimension perfectly substitutable
- B. Penalises unbalanced achievement more than a simple arithmetic mean
- C. Adjusts each dimension for inequality
- D. Converts GNI into PPP dollars

Answer: B.

Option-specific explanations:

- **A - Incorrect:** A geometric mean limits, rather than creates, perfect substitution.
- **B - Correct:** A low dimension reduces the product and therefore the composite more strongly.
- **C - Incorrect:** Dimension-level inequality discounting belongs to IHDI.
- **D - Incorrect:** PPP conversion precedes aggregation and is not performed by the mean.

Examiner trap 10: Geometric aggregation reduces but does not abolish substitutability.

MCQ 11

A country improves its HDI value but falls in rank. Which inference is valid?

- A. Its human development necessarily deteriorated
- B. UNDP changed HDI into MPI
- C. Other countries or the comparison set may have changed relatively faster
- D. Its IHDI must have risen

Answer: C.

Option-specific explanations:

- **A - Incorrect:** A higher comparable value indicates improvement, subject to revision, even if rank falls.
- **B - Incorrect:** HDI and MPI remain distinct methods.
- **C - Correct:** Rank is relative and can move differently from the country's value.
- **D - Incorrect:** HDI movement does not determine IHDI movement.

Examiner trap 11: Rank is ordinal and depends on peers, coverage and revisions.

MCQ 12

Which statement about report year and data year is correct?

- A. They are always identical
- B. A rank has no publication vintage
- C. Every component necessarily uses the same survey date
- D. A report may publish values mainly referring to an earlier reference year, which must be stated

Answer: D.

Option-specific explanations:

- **A - Incorrect:** International reports commonly use lagged inputs.
- **B - Incorrect:** Rank is tied to a specific release and comparison set.
- **C - Incorrect:** Component sources can have different vintages.
- **D - Correct:** Separating publication and reference years prevents false currency.

Examiner trap 12: Never call a report-year label the observation year without checking metadata.

MCQ 13

What is the IHDI?

- A. HDI after discounting each dimension for inequality
- B. HDI plus a fourth inequality dimension
- C. MPI divided by HDI
- D. A gender-only version of HDI

Answer: A.

Option-specific explanations:

- **A - Correct:** IHDI adjusts health, education and income indices for their distribution.
- **B - Incorrect:** Inequality is an adjustment, not an added dimension.
- **C - Incorrect:** MPI and HDI use different architectures and cannot form this ratio.
- **D - Incorrect:** GDI and GII are gender-focused measures.

Examiner trap 13: IHDI retains exactly the three HDI dimensions.

MCQ 14

When does IHDI equal HDI?

- A. When income inequality alone is zero
- B. When there is no inequality across people in all three dimensions
- C. When MPI is zero
- D. When a country has very high HDI

Answer: B.

Option-specific explanations:

- **A - Incorrect:** Equality in one dimension is insufficient if others are unequal.
- **B - Correct:** With every dimension loss zero, adjusted and unadjusted indices coincide.
- **C - Incorrect:** MPI zero does not mathematically determine IHDI.
- **D - Incorrect:** A high average can coexist with large inequality.

Examiner trap 14: The equality case is distributional, not a development-category threshold.

MCQ 15

The proportional loss due to inequality is expressed as

- A. HDI minus MPI
- B. IHDI divided by MPI
- C. 1 minus IHDI divided by HDI
- D. HDI rank minus IHDI rank

Answer: C.

Option-specific explanations:

- **A - Incorrect:** MPI is not part of the IHDI loss formula.
- **B - Incorrect:** The ratio has no such interpretation.
- **C - Correct:** The difference relative to HDI gives the proportional loss.
- **D - Incorrect:** Rank difference is ordinal and is calculated on a restricted comparison set.

Examiner trap 15: Do not confuse value loss with rank difference.

MCQ 16

Which limitation of IHDI is stated by UNDP?

- A. It omits income completely
- B. It has no health dimension
- C. It is always greater than HDI
- D. It is not association-sensitive because dimensions are not jointly observed for each person in one harmonised source

Answer: D.

Option-specific explanations:

- **A - Incorrect:** Income is one of the adjusted dimensions.
- **B - Incorrect:** Health remains a core dimension.
- **C - Incorrect:** IHDI cannot exceed HDI under the adjustment.
- **D - Correct:** The data architecture prevents observing overlapping inequalities for the same individuals.

Examiner trap 16: IHDI distribution sensitivity is not the same as MPI-style joint deprivation identification.

MCQ 17

Which list contains the global MPI's health indicators?

- A. Nutrition and child mortality
- B. Nutrition, maternal health and bank account
- C. Life expectancy and morbidity
- D. Infant mortality and insurance

Answer: A.

Option-specific explanations:

- **A - Correct:** The global method assigns nutrition and child mortality weight 1/6 each.
- **B - Incorrect:** Maternal health and bank account are national-India adaptations in different dimensions.
- **C - Incorrect:** Life expectancy belongs to HDI, and morbidity is not a named global MPI indicator.
- **D - Incorrect:** The named pair is not the global architecture.

Examiner trap 17: Do not import India's added indicators into the global MPI.

MCQ 18

In the global MPI, each living-standard indicator carries what weight?

- A. One-sixth
- B. One-eighteenth
- C. One-third
- D. One-tenth

Answer: B.

Option-specific explanations:

- **A - Incorrect:** One-sixth applies to each health and education indicator.
- **B - Correct:** Six indicators divide the one-third living-standard dimension equally.
- **C - Incorrect:** One-third is the total dimension weight.
- **D - Incorrect:** Ten indicators are not equally weighted.

Examiner trap 18: Equal dimensions do not imply equal indicator weights.

MCQ 19

Under the current global methodology, who is identified as MPI poor?

- A. Anyone deprived in one indicator
- B. Anyone below a monetary poverty line
- C. A person in a household deprived in at least one-third of weighted indicators
- D. Only a person deprived in all three dimensions

Answer: C.

Option-specific explanations:

- **A - Incorrect:** One low-weight indicator may be below the poverty cutoff.
- **B - Incorrect:** MPI identification is non-monetary.
- **C - Correct:** The weighted score must reach at least one-third.
- **D - Incorrect:** Deprivation need not include every dimension.

Examiner trap 19: The second cutoff applies to the weighted sum, not the raw number of indicators.

MCQ 20

Which classification is correctly matched in the current global method?

- A. Vulnerable means score below 10 percent
- B. Severe means score above one-third
- C. Vulnerable and severe use the same cutoff
- D. Vulnerable: 20 percent to below one-third; severe: at least 50 percent

Answer: D.

Option-specific explanations:

- **A - Incorrect:** The vulnerable band begins at one-fifth.
- **B - Incorrect:** Severe poverty uses a higher one-half threshold.
- **C - Incorrect:** The categories are distinct.
- **D - Correct:** These are the verified current global thresholds.

Examiner trap 20: Thresholds belong to the global method and should not be transferred automatically.

MCQ 21

In MPI notation, incidence H means

- A. The share of the population identified as multidimensionally poor
- B. The average deprivation score among all households
- C. The number of indicators in the index
- D. The HDI loss due to inequality

Answer: A.

Option-specific explanations:

- **A - Correct:** H equals q divided by n after applying the poverty cutoff.
- **B - Incorrect:** A, not H, is the poor's average intensity; averaging all households changes the concept.
- **C - Incorrect:** Indicator count is part of architecture, not incidence.
- **D - Incorrect:** HDI loss belongs to IHDI.

Examiner trap 21: Monetary headcount and MPI incidence use different identification rules.

MCQ 22

MPI intensity A is

- A. The poverty cutoff
- B. The average weighted deprivation share among the multidimensionally poor
- C. The national poverty headcount
- D. The sum of all uncensored deprivations in the population

Answer: B.

Option-specific explanations:

- **A - Incorrect:** The cutoff identifies poor households but is not A.
- **B - Correct:** A averages censored deprivation scores over poor people.
- **C - Incorrect:** H is the headcount ratio.
- **D - Incorrect:** Aggregation over the full population corresponds to H times A, not A alone.

Examiner trap 22: The denominator for A is the poor population.

MCQ 23

In a population, H is 0.40 and A is 0.50. The MPI is

- A. 0.90
- B. 0.45
- C. 0.20
- D. 0.10

Answer: C.

Option-specific explanations:

- **A - Incorrect:** Adding H and A is not the formula.
- **B - Incorrect:** A simple average is not the adjusted headcount.
- **C - Correct:** Multiplication gives 0.40 x 0.50 = 0.20.
- **D - Incorrect:** 0.10 would understate the product.

Examiner trap 23: MPI is a product, so preserve decimal or percentage units consistently.

MCQ 24

If H falls while A among the remaining poor rises, what can be concluded?

- A. Every poor household improved
- B. Poverty necessarily worsened
- C. The MPI must be unchanged
- D. Fewer people are identified as poor, but those remaining may face more intense deprivation

Answer: D.

Option-specific explanations:

- **A - Incorrect:** Cross-sectional H does not prove every household's trajectory.
- **B - Incorrect:** The net MPI effect depends on the magnitude of both changes.
- **C - Incorrect:** The product can rise, fall or stay constant.
- **D - Correct:** The two components can legitimately move in opposite directions.

Examiner trap 24: Report both H and A instead of narrating headcount alone.

MCQ 25

Which institution publishes India's National MPI?

- A. NITI Aayog
- B. MoSPI alone
- C. Reserve Bank of India
- D. Finance Commission

Answer: A.

Option-specific explanations:

- **A - Correct:** NITI Aayog is the nodal publisher for the national MPI exercise.
- **B - Incorrect:** MoSPI supplies important statistics but is not the publisher of this report.
- **C - Incorrect:** RBI does not own the national MPI.
- **D - Incorrect:** The Finance Commission does not publish it.

Examiner trap 25: Publisher, survey producer and technical partner are different roles.

MCQ 26

Which source periods underpin NITI Aayog's 2023 Progress Review comparison?

- A. Census 2001 and 2011
- B. NFHS-4 (2015-16) and NFHS-5 (2019-21)
- C. PLFS 2021-22 and 2022-23
- D. HDR 2022 and HDR 2023

Answer: B.

Option-specific explanations:

- **A - Incorrect:** The report is based on household health-survey rounds, not censuses.
- **B - Correct:** These are the two survey periods named in the report.
- **C - Incorrect:** PLFS is a labour-force survey and not the stated MPI source.
- **D - Incorrect:** UNDP HDRs do not supply the national comparison microdata.

Examiner trap 26: The report's 2023 release date is not its data period.

MCQ 27

Which pair is added in India's national architecture relative to the global ten-indicator list?

- A. Life expectancy and GNI per capita
- B. Political participation and safety
- C. Maternal health and bank account
- D. Employment and consumption

Answer: C.

Option-specific explanations:

- **A - Incorrect:** These are HDI indicators.
- **B - Incorrect:** Neither is in the national 12-indicator architecture.
- **C - Correct:** The national adaptation includes maternal health and bank-account access.
- **D - Incorrect:** Employment and consumption are important but not the named added indicators.

Examiner trap 27: National adaptation changes both the count and some weights.

MCQ 28

Which national-MPI weighting statement is correct?

- A. All 12 indicators receive one-twelfth
- B. Health indicators all receive one-ninth
- C. Living-standard indicators retain one-eighteenth despite adding bank account
- D. Nutrition is 1/6; child/adolescent mortality and maternal health are 1/12 each; seven living-standard indicators are 1/21 each

Answer: D.

Option-specific explanations:

- **A - Incorrect:** Equal indicator weights would not preserve the reported architecture.
- **B - Incorrect:** Health is not split into three equal weights.
- **C - Incorrect:** Adding a seventh indicator requires the stated one-third dimension to be redistributed.
- **D - Correct:** This matches the official 2023 national method.

Examiner trap 28: Do not assume equal-within-dimension weights for India's health dimension.

MCQ 29

What did NITI Aayog's 2023 report state for India's MPI headcount comparison?

- A. 24.85 percent in NFHS-4 to 14.96 percent in NFHS-5
- B. 14.96 percent in 2023 to 11.28 percent in 2026
- C. 55.3 percent in Census 2011 to 5.3 percent in PLFS 2023
- D. A current annual estimate with no survey lag

Answer: A.

Option-specific explanations:

- **A - Correct:** The report ties these values to 2015-16 and 2019-21 survey periods.
- **B - Incorrect:** The first value is not a 2023 observation, and the second pair is not the report comparison.
- **C - Incorrect:** These numbers mix unrelated vintages and measures.
- **D - Incorrect:** NFHS-based MPI is not an annual current series.

Examiner trap 29: Always attach 24.85 and 14.96 to their NFHS periods and the 17 July 2023 report.

MCQ 30

Which limitation follows from household-level MPI identification?

- A. It cannot be decomposed by indicator
- B. It may conceal unequal deprivation among members of the same household
- C. It uses only income
- D. It has no poverty cutoff

Answer: B.

Option-specific explanations:

- **A - Incorrect:** MPI is designed for indicator decomposition.
- **B - Correct:** Assigning household status to members can hide gender, age or disability differences within the household.
- **C - Incorrect:** MPI is explicitly multidimensional.
- **D - Incorrect:** The poverty cutoff is central to identification.

Examiner trap 30: Household targeting convenience comes with intra-household blindness.

MCQ 31

Which inference from rising social-service expenditure is strongest?

- A. Inclusive growth is proven
- B. HDI must rise by the same percentage
- C. The spending creates a potential conversion channel whose access, quality and outcomes must be verified
- D. Inequality has been eliminated

Answer: C.

Option-specific explanations:

- **A - Incorrect:** An input increase alone cannot prove distributional outcomes.
- **B - Incorrect:** HDI does not move mechanically with expenditure.
- **C - Correct:** The chain from allocation to capability requires utilisation and quality evidence.
- **D - Incorrect:** No expenditure series establishes equality by itself.

Examiner trap 31: Budgetary input, service output and human outcome are separate stages.

MCQ 32

Which conclusion best integrates growth, HDI, IHDI and MPI?

- A. Replace GDP entirely with MPI
- B. Use HDI rank as the sole target
- C. Treat every index change as causal proof
- D. Use real growth and jobs for resources, HDI for average achievement, IHDI for distribution and MPI for overlapping deprivation, with dated disaggregation

Answer: D.

Option-specific explanations:

- **A - Incorrect:** Production measurement remains necessary.
- **B - Incorrect:** Rank suppresses components and relativity.
- **C - Incorrect:** Index movement alone does not identify causes.
- **D - Correct:** The dashboard matches each measure to its purpose and preserves metadata.

Examiner trap 32: The correct answer is a coordinated dashboard, not a contest for one supreme index.

PYQS AND ANSWER PRACTICE

VERIFIED PYQ ROUTES AND KEY DISCIPLINE

The following demands are verified through repository routing and locally held official papers where available. No final official/local answer key or official model answer is available for these entries, so no answer is inferred.

PYQ 1 - UPSC Prelims GS-I 2018, Q48

Verified routed demand: Conditions under which rise in GNP per capita does not connote economic development.

Answer withheld pending official UPSC key.

PYQ 2 - UPSC Prelims GS-I 2019, Q77

Verified routed demand: Components of the then World Bank Ease of Doing Business index; used here only as an index-ownership and scope distinction.

Answer withheld pending official UPSC key.

PYQ 3 - UPSC Prelims GS-I 2019, Q80

Verified routed demand: Meaning of social capital in national development.

Answer withheld pending official UPSC key.

PYQ 4 - UPSC Mains GS-II 2020, Q16

Verified routed demand: Analyse the incidence and intensity of poverty in comparison with income-based poverty measurement. **15 marks, 250 words.**

Answer withheld pending official UPSC key.

PYQ 5 - UPSC Mains GS-III 2024, Q1

Official locally verified wording: "Examine the pattern and trend of public expenditure on social services in the post-reforms period in India. To what extent this has been in consonance with achieving the objective of inclusive growth?" **10 marks, 150 words.**

Answer withheld pending official UPSC key.

PYQ 6 - UPSC Mains GS-III 2025, Q1

Official locally verified wording: "Distinguish between the Human Development Index (HDI) and the Inequality-adjusted Human Development Index (IHDI) with special reference to India. Why is the IHDI considered a better indicator of inclusive growth?" **10 marks, 150 words.**

Answer withheld pending official UPSC key.

PYQ 7 - UPSC Prelims GS-I 2026, Q100

Verified routed demand: Global MPI methodology, indicators and institutional comparison. The locally held key is provisional, not final.

Answer withheld pending official UPSC key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish economic growth from economic development. Answer in 150 words.

Model answer:

Economic growth is the sustained expansion of real output or real income; development is the wider improvement in productive structure, distribution, capabilities and sustainability.

First, growth is quantitative. Real GDP, real GVA and real per-capita income show the size and pace of expansion. Development asks whether workers move into higher-productivity and decent employment. Second, Amartya Sen's capability approach shifts attention from income possessed to real freedom to be healthy, educated and secure. Third, distribution matters: a per-capita average can rise while regional, gender or social-group gaps widen. Fourth, environmental depletion can raise current output while weakening future welfare.

Growth nevertheless remains essential because jobs, household income and tax revenue finance capability formation. The correct relationship is therefore conditional: productivity-led, employment-intensive and ecologically resilient growth becomes development when institutions and public services convert resources into widely shared freedoms.

Answer method audit: Claim -> named evidence -> analysis -> qualification -> reasoned verdict.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain why the geometric mean and logarithmic income transformation are used in HDI. Answer in 150 words.

Model answer:

UNDP's HDI combines health, education and living-standard indices.

The geometric mean, `HDI = (Ih x Ie x Ii)^(1/3)`, penalises imbalance. A very high income index cannot fully compensate for weak longevity or schooling as easily as under a simple arithmetic average. It therefore reflects the proposition that all three capabilities matter.

The income index uses natural logarithms of GNI per capita. UNDP's 2025 Technical Note 1 explains the concave conversion of income into capabilities: an additional dollar normally expands choices more near subsistence than at a high income. Log transformation captures this diminishing contribution.

These choices improve conceptual discipline but do not make HDI a complete welfare measure. The index remains sensitive to goalposts and data revisions, and it omits distribution, poverty, security and environmental pressure. Hence HDI should be read with components, IHDI and other dashboards.

Answer method audit: Claim -> named evidence -> analysis -> qualification -> reasoned verdict.

ORIGINAL MAINS 3 - 15 MARKS

Question: Discuss the drivers of economic growth and the conditions under which growth becomes inclusive and sustainable development. Answer in 250 words.

Model answer:

Growth arises from labour, physical capital, human capital, technology and institutions. Capital accumulation expands capacity; education, health and skills improve labour quality; innovation and infrastructure raise productivity; and predictable institutions coordinate investment. Sustained per-capita growth ultimately requires productivity because factor accumulation faces diminishing returns.

The development outcome depends on transmission. Employment-intensive structural transformation must move workers from low-productivity vulnerability to more productive, secure work. Jobs and wages strengthen household nutrition and schooling, while profits and a broader tax base finance investment and public services. The *Economic Survey 2025-26* treats education, health, skills and employment as connected inputs into productivity and social mobility.

Three failures can break this chain. First, capital-intensive enclaves may generate job-poor growth. Second, inequality and weak services may prevent poorer households from converting income into capabilities. Third, pollution, resource depletion and climate shocks can shift costs to vulnerable groups and future generations.

Policy should therefore combine competitive investment and infrastructure with labour-intensive sectors, female participation, quality health and education, portable social protection, lagging-region investment and a just green transition. Outcomes, not expenditure alone, should be tracked through jobs, real wages, learning, health, HDI/IHDI, monetary poverty and MPI.

Thus growth is the resource base, but inclusion, capability conversion and ecological resilience determine whether it becomes durable development.

Answer method audit: Claim -> named evidence -> analysis -> qualification -> reasoned verdict.

ORIGINAL MAINS 4 - 15 MARKS

Question: Compare HDI and IHDI and assess why IHDI is more relevant to inclusive growth. Answer in 250 words.

Model answer:

HDI and IHDI share the same three dimensions - health, education and standard of living - but answer different questions.

HDI measures average achievement. It normalises life expectancy, expected and mean years of schooling, and GNI per capita in PPP terms, then takes the geometric mean of the three dimension indices. It can compare capability levels and expose why similar incomes yield different social outcomes.

IHDI adjusts each dimension for inequality. Under UNDP's 2025 Technical Note 2, an Atkinson-type inequality measure discounts the health, education and income indices; their geometric mean forms IHDI. With perfect equality IHDI equals HDI; as inequality rises IHDI falls below it. The proportional loss is `1 - IHDI/HDI`.

IHDI is therefore more relevant to inclusive growth because it tests whether average achievement is broadly distributed. Two countries with the same HDI may have different IHDIs when longevity, schooling or income is concentrated.

However, IHDI is not a fourth dimension, a poverty rate or a complete inclusion index. UNDP notes that it is not association-sensitive because all dimensions are not jointly observed for each individual. It also omits employment quality, voice and environment.

For India, HDI should establish average capability and IHDI the distributional discount, supplemented by MPI, jobs and subgroup dashboards.

Answer method audit: Claim -> named evidence -> analysis -> qualification -> reasoned verdict.

ORIGINAL MAINS 5 - 20 MARKS

Question: Explain the global MPI methodology and evaluate its advantages and limitations relative to income poverty. Answer in 250 words.

Model answer:

The global Multidimensional Poverty Index, developed by OPHI with UNDP, measures acute overlapping deprivation through the Alkire-Foster method.

Its three equally weighted dimensions are health, education and living standards. Nutrition, child mortality, years of schooling and school attendance each weigh one-sixth. Cooking fuel, sanitation, drinking water, electricity, housing and assets each weigh one-eighteenth. Indicator-specific cutoffs identify each household deprivation; weights are summed into score `c`. A person in a household with `c >= 1/3` is MPI poor. Incidence `H` is the poor population share; intensity `A` is their average deprivation score; `MPI = H x A`.

MPI improves on a monetary headcount in three ways. It reveals deprivation composition and overlap; it distinguishes how many are poor from how poor they are; and it can be decomposed by indicator, region and group for targeting. Public services may also improve direct outcomes without an immediate equivalent rise in household income.

Yet MPI is not superior for every purpose. Indicator, weight and cutoff choices affect results. Household identification can hide intra-household inequality. Survey lag misses sudden shocks, and threshold crossing can create classification sensitivity. MPI does not estimate the cash shortfall required to cross a monetary line.

Policy should therefore use income poverty for resource adequacy and MPI for direct, overlapping capability deficits, supported by uncensored indicators and sensitivity analysis.

Answer method audit: Claim -> named evidence -> analysis -> qualification -> reasoned verdict.

ORIGINAL MAINS 6 - 20 MARKS

Question: Assess India's National MPI as a tool for federal and district development policy. Answer in 250 words.

Model answer:

India's National MPI translates the Alkire-Foster framework into a domestic diagnostic led by NITI Aayog and based principally on NFHS household microdata.

The index uses health, education and standard-of-living dimensions but adapts the global architecture to 12 indicators. It adds maternal health and bank account. Nutrition weighs one-sixth; child/adolescent mortality and maternal health one-twelfth each; the two education indicators one-sixth each; and seven living-standard indicators one-twenty-first each.

Its principal strength is decomposability. The 2023 Progress Review compares NFHS-4 (2015-16) and NFHS-5 (2019-21), reports the national headcount falling from 24.85 to 14.96 percent, and provides State/UT and district evidence. Governments can identify whether nutrition, schooling, sanitation, housing or finance drives a local burden, assign departmental responsibility and target lagging districts.

However, the report's publication date is not the data year. NFHS periodicity creates lag; repeated cross-sections do not track the same households; district averages can hide social-group differences; and household classification can conceal unequal deprivation within families. National indicator definitions and weights also mean its values are not interchangeable with the global MPI.

The tool should guide cooperative federalism through transparent district dashboards, outcome audits and convergence of health, education, nutrition and basic-service programmes, while monetary poverty, employment and individual-level gender evidence remain complementary.

Answer method audit: Claim -> named evidence -> analysis -> qualification -> reasoned verdict.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

OPTIONAL ADVANCED 1 - Why geometric aggregation matters

The geometric mean reduces perfect substitutability across HDI dimensions: a weak dimension pulls the product down more strongly than under a simple arithmetic mean. It does not eliminate substitutability, because improvement in one dimension can still offset some weakness elsewhere. The education dimension itself uses an arithmetic average of the EYS and MYS indices under the 2025 method.

OPTIONAL ADVANCED 2 - IHDI's distributional method and association limit

The 2025 IHDI uses Atkinson-type dimension losses with epsilon equal to one. Its subgroup-consistency property is useful, but UNDP explicitly notes that it is not association-sensitive: health, schooling and income inequalities are not observed jointly for every person from one harmonised global survey. IHDI therefore cannot identify the same people as simultaneously deprived across dimensions.

OPTIONAL ADVANCED 3 - Alkire-Foster robustness and governance

The Alkire-Foster method is decomposable by subgroup and indicator. Responsible use should test whether broad conclusions survive reasonable changes in weights and cutoffs. Indicator improvement may arise through genuine capability gain, reporting change or movement just across a threshold; dashboards should preserve uncensored indicator information.

OPTIONAL ADVANCED 4 - GDI and GII as bounded comparisons

GDI compares female and male HDI achievement in the same three dimensions using sex-disaggregated indicators. GII measures gender disadvantage through reproductive health, empowerment and labour-market dimensions. Neither is an extra HDI dimension; both belong to gender-focused analysis and should be used only to expose a limitation of aggregate HDI here.

OPTIONAL ADVANCED 5 - Sustainable and inclusive measurement

Development evaluation can supplement production, employment, HDI/IHDI and poverty with emissions, material use, natural wealth, vulnerability and resilience. This does not authorise an improvised mega-index. A transparent dashboard usually preserves causal diagnosis better than compressing every concern into one number.

CONSOLIDATED REGISTER NOTES

1. Growth-development distinction

- Growth = sustained quantitative rise in **real** output/income; per-capita growth adjusts for population but not distribution.
- Development = growth plus structural transformation, capabilities, distribution, agency, resilience and sustainability.
- Drivers: labour, physical capital, human capital, technology, infrastructure and institutions.
- Transmission: productivity -> jobs/wages/profits/tax base -> household and public capability investment.
- Failures: jobless growth, regional concentration, unequal access, weak service quality and environmental externalities.

2. HDI rapid decoder

Dimension	Indicator	2025 Technical Note goalposts
Long and healthy life	Life expectancy at birth	20-85 years
Knowledge	Expected years of schooling	0-18 years
Knowledge	Mean years of schooling, adults 25+	0-15 years
Decent standard of living	GNI per capita	100-75,000 constant 2021 PPP dollars

- `Education index = (EYS index + MYS index)/2`.
- `Income index = [ln(actual)-ln(100)]/[ln(75,000)-ln(100)]`.
- `HDI = (Health x Education x Income)^(1/3)`.
- Geometric mean limits compensation across dimensions; logarithmic income reflects diminishing capability gains.
- Rank is relative; value is the composite score; report year and data year must be stated separately.

3. IHDI rapid decoder

- Same three HDI dimensions; **not a fourth inequality dimension**.
- `I-adjusted = (1-Ax) x I`; `IHDI = (Health-adjusted x Education-adjusted x Income-adjusted)^(1/3)`.
- `Loss = 1 - IHDI/HDI`.
- Equality case: all dimension inequalities zero -> IHDI equals HDI.
- Limitation: not association-sensitive across the same individuals; smaller data coverage.

4. Global MPI rapid decoder

- OPHI + UNDP; Alkire-Foster; acute overlapping deprivation.
- Health: nutrition and child mortality, 1/6 each.
- Education: years of schooling and school attendance, 1/6 each.
- Living standards: fuel, sanitation, water, electricity, housing and assets, 1/18 each.
- Dual cutoff: indicator deprivation first; overall `c >= 1/3` identifies MPI poor.
- Current global bands checked 9 September 2026: vulnerable `1/5 <= c < 1/3`; severe `c >= 1/2`.
- `H = q/n`; `A = poor people's average weighted deprivation`; `MPI = H x A`.

5. India's National MPI rapid decoder

- Nodal publisher: NITI Aayog; household source: NFHS.
- 12 indicators: adds maternal health and bank account to the global indicator themes.
- Weights: nutrition 1/6; child/adolescent mortality 1/12; maternal health 1/12; both education indicators 1/6; seven living-standard indicators 1/21.
- 2023 Progress Review, released 17 July 2023, compares NFHS-4 (2015-16) with NFHS-5 (2019-21).
- Report facts: headcount 24.85% -> 14.96%; estimated 13.5 crore people moved out of multidimensional poverty.
- Uses: national, State/UT, district and indicator decomposition; limitation: lag and national/global non-equivalence.

6. Comparison and limitation matrix

Measure	Core question	Principal blind spot
Real GDP/per capita	How large and fast is output?	Distribution and non-market capabilities
HDI	What is average achievement?	Within-country distribution and omitted dimensions
IHDI	What remains after inequality discount?	Joint overlap across the same people
Monetary poverty	Is command over resources below a line?	Direct service/capability overlap
MPI	Who crosses a weighted deprivation cutoff and how intensely?	Indicator/cutoff choice, intra-household inequality and lag

7. Must-not-miss traps

- HDI: 3 dimensions, 4 indicators; GNIpc PPP, not GDP alone.
- Final HDI geometric mean; education subindices arithmetic mean.
- IHDI <= HDI; the gap is not a poverty rate.
- Global MPI has 10 indicators; India's national MPI has 12.
- Equal dimension weights do not imply equal weights for all indicators.
- H asks how many; A asks how deprived among the poor.
- Publication year is not data year; rank movement is not value movement.
- Expenditure is an input; verify access, quality, utilisation and outcome.

8. Mains answer spine

```
Define -> decode formula/institution -> date evidence -> explain growth/jobs/services mechanism
-> compare distribution/deprivation -> qualify averages, cutoffs, household unit, lag and
causation -> recommend productive inclusion, quality public services and ecological resilience.
```

COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM

```
PANEL 1/12 - CORE CONCEPT: GROWTH IS A MEANS, DEVELOPMENT THE TEST
+---------------------------------------------------------------------------+
| REAL GROWTH = sustained increase in inflation-adjusted output or income     |
| DEVELOPMENT = growth + structural change + distribution + capabilities     |
|               + agency + resilience + environmental sustainability         |
| REAL GDP -> aggregate quantity; REAL GDP PER CAPITA -> average quantity     |
| Neither alone proves decent jobs, equal access, security or future welfare. |
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PANEL 2/12 - GROWTH ENGINE AND CONVERSION CHAIN
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| labour + physical capital + human capital + technology + institutions       |
|                          -> productivity -> real output                      |
| output -> jobs/wages + profits/investment + tax base                        |
| jobs/services -> nutrition + health + schooling + agency -> productivity    |
| Failure modes: jobless growth, enclave growth, inequality, weak services.   |
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PANEL 3/12 - HDI TIMELINE, PURPOSE AND FOUR INDICATORS
+---------------------------------------------------------------------------+
| 1990: first UNDP Human Development Report                                   |
| PURPOSE: average achievement, not complete welfare                          |
| HEALTH: life expectancy at birth                                             |
| EDUCATION: expected years schooling + mean years schooling (age 25+)        |
| LIVING STANDARD: GNI per capita in PPP terms                                |
| Three dimensions; four indicators; publisher = UNDP.                        |
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PANEL 4/12 - HDI DECODER, 2025 TECHNICAL NOTE
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| Linear index = (actual - minimum)/(maximum - minimum)                        |
| LE goalposts 20-85; EYS 0-18; MYS 0-15                                     |
| GNIpc goalposts 100-75,000 constant 2021 PPP dollars                        |
| Education index = arithmetic mean of EYS and MYS indices                    |
| Income index uses ln values: diminishing capability return to income        |
| HDI = (Health x Education x Income)^(1/3): geometric mean                    |
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PANEL 5/12 - HDI INTERPRETATION FIREWALL
+---------------------------------------------------------------------------+
| VALUE = composite achievement; RANK = relative order in a stated release    |
| REPORT YEAR != DATA YEAR; components may have different source vintages     |
| Rank can move because peers, coverage or revisions change.                  |
| Limits: average hides inequality; only 3 dimensions; omits poverty, voice,  |
| security, service quality, unpaid care and environmental pressure.          |
| GDI/GII are bounded gender complements, not fourth HDI dimensions.          |
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PANEL 6/12 - IHDI: SAME DIMENSIONS, DISTRIBUTION DISCOUNT
+---------------------------------------------------------------------------+
| Atkinson-type inequality Ax for health, education and income                 |
| Adjusted dimension I-adjusted = (1 - Ax) x I                                |
| IHDI = (Health-adjusted x Education-adjusted x Income-adjusted)^(1/3)       |
| Loss = 1 - IHDI/HDI; perfect equality -> IHDI = HDI; always IHDI <= HDI     |
| Not a fourth dimension; gap is not a poverty rate.                          |
| UNDP limit: not association-sensitive across the same individuals.          |
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PANEL 7/12 - GLOBAL MPI: 3 DIMENSIONS, 10 INDICATORS
+---------------------------------------------------------------------------+
| OPHI + UNDP; acute household-level overlapping deprivation                  |
| Health 1/3: nutrition 1/6; child mortality 1/6                              |
| Education 1/3: years schooling 1/6; attendance 1/6                          |
| Living standards 1/3: fuel, sanitation, water, electricity, housing, assets |
|                        each 1/18                                             |
| Equal dimension weights do not mean equal weights for all 10 indicators.    |
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PANEL 8/12 - DUAL CUTOFF AND MPI = H x A
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| 1st cutoff: indicator-specific deprivation; sum weights -> score c            |
| 2nd cutoff: c >= 1/3 -> MPI poor                                               |
| Current global bands: vulnerable 1/5 to <1/3; severe c >= 1/2                 |
| H = poor population / total population                                         |
| A = average weighted deprivation score among the poor                         |
| MPI = H x A; H and A can move in opposite directions.                         |
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PANEL 9/12 - INDIA'S NATIONAL MPI IS AN ADAPTATION
+--------------------------------------------------------------------------------+
| Publisher/nodal institution: NITI Aayog; source: NFHS household microdata    |
| 12 indicators: global spine + maternal health + bank account                  |
| Health: nutrition 1/6; mortality 1/12; maternal health 1/12                   |
| Education: MYS 1/6; attendance 1/6                                             |
| Living standards: 7 indicators, each 1/21                                     |
| Used at national, State/UT and district levels; not interchangeable globally.|
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PANEL 10/12 - DATED INDIA EVIDENCE AND DATA-LAG RULE
+--------------------------------------------------------------------------------+
| NITI Progress Review released 17 July 2023                                    |
| compares NFHS-4 (2015-16) with NFHS-5 (2019-21)                               |
| headcount ratio: 24.85% -> 14.96%; report estimate: 13.5 crore exited        |
| These are survey-period comparisons, not observations made in 2023.          |
| Limits: periodic survey, repeated cross-section, household assignment,       |
| district/social-group averages and indicator/cutoff sensitivity.             |
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PANEL 11/12 - COMPARISON, LIMITS AND PRELIMS TRAPS
+--------------------------------------------------------------------------------+
| Income poverty: monetary command below line; MPI: direct overlapping deficits|
| HDI: average achievement; IHDI: inequality-adjusted achievement               |
| Global MPI: 10 indicators; India national MPI: 12 indicators                  |
| H != A; IHDI loss != MPI; GNIpc != GDPpc; rank != value                       |
| Household MPI can hide intra-household gender/age/disability inequality.     |
| Expenditure != access != quality != utilisation != outcome.                  |
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PANEL 12/12 - ANSWER SPINE AND QUALIFIED VERDICT
+--------------------------------------------------------------------------------+
| DEFINE -> select metric -> decode institution/unit/formula/threshold         |
| -> attach report year + data year -> calculate/compare components            |
| -> explain productivity/jobs/services/distribution/environment mechanism     |
| -> qualify averages, cutoff sensitivity, household unit, lag and causation  |
| -> prescribe productive inclusion + quality capabilities + resilience        |
| VERDICT: use a dated, disaggregated dashboard; no single index is development.|
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```